1.23 e and Natural Logarithms

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$y = e^x$

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Click ↓

Function	Gradient	
$y = 1^x$		>
$y = 1.5^x$		>
$y = 2^x$		>
$y = 2.5^x$		>
$y = 3^x$		>
$y = 3.5^x$		>

Compare each exponential function against its respective gradient function. What do you notice?

You won't yet be able to differentiate general exponential functions $y = a^x$ until Year 2.
But I've calculated the gradient functions for you – click the black arrow to reveal the graph and gradient function.

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$y = e^x$

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Function	Gradient	
$y = 1^x$	$\frac{dy}{dx} = 0$	>
$y = 1.5^x$	$\frac{dy}{dx} = 0.41 \times 1.5^x$	>
$y = 2^x$	$\frac{dy}{dx} = 0.69 \times 2^x$	>
$y = 2.5^x$	$\frac{dy}{dx} = 0.92 \times 2.5^x$	>
$y = 3^x$	$\frac{dy}{dx} = 1.10 \times 3^x$	>
$y = 3.5^x$	$\frac{dy}{dx} = 1.25 \times 3.5^x$	>

$y = 2.5^x$ and $y = 3^x$ seem to be similar to their respective gradient functions. So is there a base between 2.5 and 3 where the **function is equal to its gradient function**?

$e = 2.71828...$ is known as **Euler's Number.**

It is one of the five most fundamental constants in mathematics (0, 1, i, e, π).

It has the property that:

$$y = e^x \quad \rightarrow \quad \frac{dy}{dx} = e^x$$

Although any function of the form $y = a^x$ is known as **an** exponential function, e^x is known as "**the**" exponential function.

You can find the exponential function on your calculator, to the right (above the "ln" key)

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Differentiating $y = ae^{kx}$

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If $y = e^{kx}$, where k is a constant, then $\frac{dy}{dx} = ke^{kx}$

Note: This is not a standalone rule but an application of something called the '*chain rule*', which you will encounter in Year 2.

Different e^{5x} with respect to x.

Different e^{-x} with respect to x.

$y = e^{-1x} \qquad \therefore \quad \frac{dy}{dx} =$

Different $4e^{3x}$ with respect to x.

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Natural Logarithms

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We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.
We also saw that e^x is "**the**" exponential function.
The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Solve $e^x = 5$

"ln both sides". On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.

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Quadratics in e^x

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In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation **is** then quadratic, or (strongly recommended!) **go straight for the factorisation**.

Solve $e^{2x} + 2e^x - 15 = 0$

Note that $e^{2x} = (e^x)^2$ therefore

Solve $e^x - 2e^{-x} = 1$

First write negative powers as fractions:

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Test Your Understanding

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Solve $\ln(3x + 1) = 2$

Solve $e^{2x} + 5e^x = 6$

Solve $2^x e^{x+1} = 3$ giving your answer as an exact value.

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4 $Find the exact solutions to the$ equation $e^{x} + 12e^{-x} = 7$.

6 Solve $3^{x}e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$

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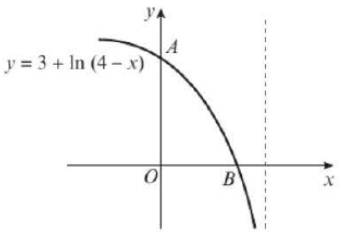
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8 The graph of $y = 3 + \ln(4 - x)$ is shown to the right.

a State the exact coordinates of point A. \hfill (1

b Calculate the exact coordinates of point B. \hfill (3 m



Challenge: The graph of the function passes through (0,5) and (6,10). Given that the line $y = 2$ is an asymptote to the graph, show that $B = \frac{1}{6}\ln\left(\frac{8}{3}\right)$